Apr. 14

- **Classification of finite Abelian Groups**

Let G be an Abelian group with order n. Then there is a factorization of n: $n = p_1^{e_1} \cdot p_2^{e_2} \cdot \ldots \cdot p_r^{e_r}$ where the p_i are primes that are not necessarily distinct such that $G \cong \mathbb{Z}/p_1^{e_1}\mathbb{Z} \times \mathbb{Z}/p_2^{e_2}\mathbb{Z} \times \ldots \times \mathbb{Z}/p_r^{e_r}\mathbb{Z}$.

Every finite Abelian group breaks down into cyclic groups of prime order

Ex. $n=16$

= same max element order (4)

= different max element order (2 vs 2^2)

$16 = 2^4$ $\quad \mathbb{Z}/2^4\mathbb{Z}$

$16 = 2 \cdot 2^3$ $\quad \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2^3\mathbb{Z}$

$16 = 2 \cdot 2 \cdot 2^2$ $\quad \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2^2\mathbb{Z}$ notice this is not same as $\mathbb{Z}/2^2\mathbb{Z} \times \mathbb{Z}/2^2\mathbb{Z}$ $\quad 16 = 2^2 \cdot 2^2$

$16 = 2 \cdot 2 \cdot 2 \cdot 2$ $\quad \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$

Ch. 12: Rings

- $\mathbb{Z}$ is one of the most important/useful rings
  - If a ring has a mult. identity, $\exists$ a homomorphism from $\mathbb{Z}$ into that ring

- **Def** A ring R (not $\mathbb{R}$, the real numbers) is a set with two binary operations, $+$ (plus/addition) and $\cdot$ (times/multiplication)

  closure is implicit $+ : R \times R \to R$, $\cdot : R \times R \to R$

  Note: this $+$ and $\cdot$ may not be exactly like regular addition and mult., depends on the ring

  conditions:
  1) $(R, +)$ is an Abelian group
  2) $(R, \cdot)$ is associative $(a \cdot b) \cdot c = a \cdot (b \cdot c)$ $\forall a,b,c \in R$
  3) Multiplication is left and right distributive over addition $a \cdot (b+c) = a \cdot b + a \cdot c$ and $(a+b) \cdot c = ac + bc$ $\forall a,b,c \in R$

- If, additionally, mult. is commutative i.e $ab = ba$ $\forall a,b \in R$ we say R is a commutative ring
- If $\exists$ a mult. identity in R, then we say R is a ring with unity or mult. identity, which is often denoted by 1 $\quad 1 \cdot a = a \cdot 1 = a$

Some definitions require 1 to be a ring, we will not, unity is more specific ring

- Typically, we denote the additive identity of a ring by 0

Examples $(\mathbb{Z}, +, \cdot)$, $(\mathbb{Z}/m\mathbb{Z}, +, \cdot)$, $(\mathbb{Q}, +, \cdot)$, $(\mathbb{R}, +, \cdot)$, $(\mathbb{C}, +, \cdot)$ are commutative rings

$(\mathbb{Z}, +, \cdot) \subseteq (\mathbb{Q}, +, \cdot) \subseteq (\mathbb{R}, +, \cdot) \subseteq (\mathbb{C}, +, \cdot)$

- $(\mathbb{N}, +, \cdot)$ is not a ring, no additive inverses
- $\mathbb{Z}[x] = \{$polynomials in the indeterminant (variable) x w/coefficients in $\mathbb{Z}\}$, $(\mathbb{Z}[x], +, \cdot)$ is a commut. ring

- $(2\mathbb{Z}, +, \cdot)$ is a commutative ring (not with unity)
- $M_{2\times 2}(\mathbb{Q}) = \{2\times 2 \text{ matrices w/ coefficients in } \mathbb{Q}\}$ is a non-commutative ring

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 \\ 2 & -1 \end{bmatrix} = \begin{bmatrix} 1+4 & 1-2 \\ 3+8 & 3-4 \end{bmatrix} = \begin{bmatrix} 5 & -1 \\ 11 & -1 \end{bmatrix}$$

with <u>matrix</u> add and mult.

under entrywise add and mult, it's a commutative ring

$$\begin{bmatrix} 1 & 1 \\ 2 & -1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 4 & 6 \\ -1 & 0 \end{bmatrix} \neq$$

not commutative

Also, with unity, mult. id is the identity matrix $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

- functions $\mathbb{R} \to \mathbb{R}$ under $+$ and $\cdot$ is a commutative ring with unity

  under composition is slightly more complicated, see matrix mult = comp. of linear transformations

- <u>Def.</u> If R is a ring with unity and $u, v \in R$ are elements such that $uv = 1$, then we say u and v are <u>units</u>
  - The units of a ring form a group, called the <u>multiplicative group of R</u>, denoted $R^\times, R^*, U(R)$

- <u>Def</u> If $a, b \in R$, then we say <u>a divides b</u> and we write $a|b$ if $a \cdot c = b$ for some $c \in R$

INTRO TO RINGS

MATH 470

A *ring* is a set R with two binary operations (denoted $+$ and $\cdot$) such that

(i) $(R, +)$ is an Abelian group.

(ii) $(R, \cdot)$ is associative.

(iii) The "multiplication operation" $\cdot$ is left and right distributive over $+$. Symbolically, $a \cdot (b + c) = a \cdot b + a \cdot c$ and $(b + c) \cdot a = b \cdot a + c \cdot a$ for all $a, b, c \in R$.

If $(R, \cdot)$ is commutative, then R is called a *commutative ring*. If R has an element 1, such that $1 \cdot a = a \cdot 1$ for all $a \in R$, then we call 1 *unity* and say R has unity. Note, we often abbreviate $a \cdot b$ by ab.

1) What is an example of an infinite noncommutative ring?

$M_{n\times n}(\mathbb{Z})$ under matrix addition and matrix multiplication

2) What is an example of a finite commutative ring with unity?

$(\mathbb{Z}/m\mathbb{Z}, +, \cdot)$

3) What is the smallest ring?

$\{0\}$, the zero ring

4) What is an example of a ring without unity?

$(2\mathbb{Z}, +, \cdot)$

5) What is an example of a finite noncommutative ring?

$M_{n\times n}(\mathbb{Z}/m\mathbb{Z})$ under matrix addition and matrix multiplication

6) Let R and S be two rings. Give a proof sketch that the direct product $R \times S$ with component-wise operations is a ring.

$R\times S = \{(r,s) : r\in R, s\in S\}$, define $+$ by $(r_1,s_1)+(r_2,s_2) = (r_1 +_r r_2, s_1 +_s s_2)$ and $\cdot$ by $(r_1,s_1)\cdot(r_2,s_2) = (r_1 \cdot_r r_2, s_1 \cdot_s s_2)$. Show $R\times S$ forms Abelian group under $+$ (group axioms, and $(r_1,s_1)+(r_2,s_2) = (r_2,s_2)+(r_1,s_1)$), associative under $\cdot$ i.e $(r_1,s_1)\cdot((r_2,s_2)\cdot(r_3,s_3)) = ((r_1,s_1)\cdot(r_2,s_2))\cdot(r_3,s_3)$, and distributivity holds (I'm not writing all that out but you get the idea, it's a sketch)

$GL_2(\mathbb{R})$ is not closed under + either, $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

7) Is $GL_2(\mathbb{R})$ with matrix addition and multiplication a ring? Is $\mathbb{R}^\times$ with multiplication and exponentiation a ring?

- $GL_2(\mathbb{R})$ is not, because it does not contain the zero matrix $\det \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = 0$, so it is not invertible, therefore not in $GL_2(\mathbb{R})$

8) Prove $a \cdot 0 = 0 \cdot a = 0$ for all $a \in R$.

9) Prove $a \cdot (-b) = (-a) \cdot b = -ab$ for all $ab \in R$. Further, show $(-a) \cdot (-b) = ab$.

10) Prove that if a ring has unity, then it is unique. Prove that if an element $a \in R$ has a multiplicative inverse, then it is unique.

11) Prove that if $0 = 1$ in a ring R, then $R = \{0\}$. We call $\{0\}$ the *zero ring*.